



Passionate and aspiring DevOps Engineer with a strong foundation in Python, cloud automation, and CI/CD pipelines. Skilled in Docker and Git, with hands-on experience in deploying scalable systems and optimizing workflows. Eager to apply my skills and customer-focused insights to drive innovative solutions in a DevOps role.

✉ konnect.surabhi@gmail.com

🏠 Ritterstr. 16, 09111

🌐 <https://sshsurabhi.netlify.app/>

🇮🇳 Indian

📍 Chemnitz

📞 0176 83334794

💍 Married

Social networks

🌐 @sshsurabhi

🐦 @sshsurabhi

👤 @SaiSriHarsha

💬 @live:surabhi.harsha

Languages

German

B2

English

C1

Telugu

Native [Mother Tongue]

Hindi

C1

Skills

AWS-DevOps | Python | Terraform | Kubernetes | Docker | Ansible | Jenkins | Gitops | Linux | BASH | PostgreSQL | MongoDB | Automation | Testing | Image Processing | PyQt | Machine Learning | Django | HTML | CSS | MATLAB | React | REST API | C++ |

Work experience

Development Engineer

06/2021 - 06/2024

Moewe Optical Solutions, Mittweida, SN, Germany

- Developed an algorithm for error detection utilizing various image processing techniques.
- Created a cross-functional desktop application using PyQt to support laser projection functionalities.
- Updated project features with new testing capabilities in accordance with customer requirements.
- Designed function-based test scripts to ensure smooth communication processes.
- Worked with communication protocols such as I2C, UART, TCP/IP, UDP, & Ethernet.
- Implemented regression testing to enhance workflow stability and maintain software integrity.
- Developed automated testing scripts (pytest) to improve efficiency and accuracy.
- Conducted unit and functional tests to ensure system robustness.
- Developed a Petalinux environment for effective communication with FPGA.
- Created Linux daemons to optimize overall system functionality.
- Application containerizing using Docker to simplify deployment and usage.
- Utilized Git for version control, collaboration, code management, and tracking changes throughout the development.

Student Research Assistant (HI-Wi)

04/2020 - 09/2020

Technische Universität Chemnitz, Chemnitz, SN, Germany

- Conducted comprehensive testing and evaluation of 3D hand pose estimation algorithms for accuracy and performance.
- Utilized the VGG annotator to create precise ground-truth masks for hands, enhancing dataset reliability.
- Tested the hand pose estimation pipeline with diverse datasets, validating outputs against established ground truth values.
- Investigated methods to reduce message data size, contributing to more efficient data transmission.
- Evaluated pixel (2D) and positional (3D) accuracy to ensure the reliability of hand pose estimation techniques.

Software Test Engineer

10/2012 - 09/2014

Caagnota Info Pvt. Ltd., Hyderabad, India

- Created and executed comprehensive test cases for software function blocks, ensuring thorough validation of system functionalities.
- Led the execution of Functional Testing, System Integration Testing, and Regression Testing, contributing to overall software quality assurance.
- Analyzed and documented error reports for failed test cases, facilitating prompt resolution and continuous improvement.
- Coordinated team activities and managed project workflows, fostering effective collaboration within Agile frameworks.
- Worked within Agile testing methods, adapting to dynamic project requirements and ensuring timely deliverables.

Education

Diploma in Devops Engineer

06/2024 - 10/2024

Datascientest [Sorbonne University], Paris, France

Integrationskurs Deutsch

11/2020 - 04/2021

Initiativ Gruppe, München, BY, Germany

Masters in Information and Communication Systems

10/2014 - 09/2019

TU Chemnitz, Chemnitz, SN, Germany

Masters in Electronics

08/2010 - 06/2012

Osmania University, Hyderabad, TG, India

Skills

Network Protocols
TCP/IP, UDP, ETHERNET

Software Programming
Python, Matlab, HTML, CSS, C# (Basic), C/C++ (Basic)

Communication Protocols
CAN, LIN

Technical Tools

Editors
Visualstudio Code, Visualstudio, PyCharm, Wireshark, GitHub, Gitlab, Jira, Trello, Confluence, Arduino, Putty, MicrochipStudio, Docker, H-Term

Certifications

DevOps, Ansible, Terraform, Kubernetes, Jenkins, Docker, SQL, NoSQL, Gitlab, Linux-Administration, AWS Solution Architect [Udemy], Deutsch, ISTQB [Udemy]
[\[Link for certifications\]](#)

Other Skills

Backend Programming

- Django [RestAPI]
- FastAPI
- Flask

Manual And Autoimation Testing with Python

- GUI Tesing
- API Testing
- Manual Testing
- Electronic Board level Testing

Operating Systems

Windows [10, 11]

Linux [Ubuntu, Fedora, Scientific Linux]

Interests

Hiking/Trekking, Cycling.
Learning New Technologies

Projects

Devops Project [Deploying Spring-PetClinic in AWS Cloud] 06/2024 - 10/2024
Datscientest.com, Paris, France

- Topic:** Deploying Spring PetClinic Project in AWS Cloud using Microservices. [\[Git Link\]](#)
- Deployed and managed a scalable, high-availability infrastructure on AWS, leveraging services such as EC2, S3, RDS, Route S3, and Lambda.
 - Automated the provisioning of cloud resources using Terraform and AWS CloudFormation, significantly reducing deployment times and streamlining setup processes.
 - Designed and implemented CI/CD pipelines using Jenkins, GitLab CI, and AgroCD to automate the testing, building, and deployment of applications in AWS.
 - Managed containerized applications using Docker and Kubernetes (EKS).
 - Improved application security by implementing IAM roles, configuring security groups, and utilizing AWS Systems Manager for automated security patching.
 - Developed and maintained comprehensive monitoring and alerting solutions using AWS CloudWatch, Prometheus, and Grafana to ensure system reliability and quick incident response.
 - Led cloud migration projects and implemented effective disaster recovery solutions across multiple AWS regions, significantly reducing downtime and minimizing risks of data loss.
 - Built CI/CD pipelines in Jenkins, incorporating unit testing to ensure code quality and reliability.

Terraform [Deploying highly available application in AWS] 09/2024 - 10/2024
Datscientest.com, Paris, France

- Designed and deployed a highly available and scalable application using Terraform, leveraging modular and reusable code to ensure repeatability and efficiency. [\[Git Link\]](#)
- Configured a secure AWS environment with 1 VPC, multiple subnets (private and public), and NAT gateways to facilitate internet access for private resources.
- Implemented auto-scaling groups for EC2 instances hosting the application, with a minimum of 1 instance and a maximum of 2 to ensure performance and availability. Deployed AWS RDS across multiple availability zones to achieve database redundancy and reliability.
- Configured an Application Load Balancer (ALB) to evenly distribute traffic, ensure fault tolerance, and enable secure HTTPS (TLS) access for web servers.
- Deployed a bastion host in an auto-scaling group to provide a secure and scalable entry point for managing private resources.

Kubernetes Project 08/2024
Datscientest, Paris, France

- Designed and deployed Kubernetes-based microservices: FastAPI, PostgreSQL, and PGAdmin, with scalable and secure configurations. [\[Github Link\]](#)
- Implemented deployments using YAML, Helm charts, and Kustomize, ensuring modularity and reusability.
- Configured HorizontalPodAutoscaler (HPA) to scale pods based on CPU utilization and provisioned a shared 10 Gi PersistentVolume for storage.
- Secured the application with SSL certificates via Let's Encrypt and exposed it through an Ingress controller.
- Prepared a disaster recovery plan using the K3s client and implemented ETCD database backups.

Instance Segmentation for Human Localization [Master Thesis] 04/2019 - 09/2019
Technische Universität Chemnitz, Chemnitz, SN, Germany

- **Architecture:** Designed and implemented a robust instance segmentation architecture specifically aimed at accurately detecting and localizing individuals, even in challenging scenarios where occlusions occur due to other people or objects.
- **Annotation:** Utilized the COCO Annotator to meticulously annotate disjoint body regions, enabling precise segmentation of individuals and improving the quality of training data.
- **Advanced Techniques:** Applied sophisticated recognition methods to enhance the identification of human figures within crowded environments.
- **Optimization:** Conducted experiments with various algorithms to achieve optimal accuracy, refining the model's performance through iterative testing and validation.
- **Validation and Testing:** Tested the segmentation algorithm to ensure effective recognition of actual human regions against original masks, validating its efficiency in practical applications and real-world scenarios.